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DESIGN AND ANALYSIS OF ALGORITHMS

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ANALYSIS OF RECURSIVE ALGORITHMS



Recurrence Relation

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- A recurrence relation, $T(n)$, is a recursive function of an integer variable n .
- Like all recursive functions, it has one or more recursive cases and one or more base cases.

For example,

$$T(n) = \begin{cases} a & \text{if } n = 1 \\ 2T(n/2) + bn + c & \text{if } n > 1 \end{cases}$$

Recurrence Relation

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$$T(n) = \begin{cases} a & \text{if } n = 1 \\ 2T(n/2) + bn + c & \text{if } n > 1 \end{cases}$$

- The portion of the definition that does not contain T is called the base case of the recurrence relation;
- the portion that contains T is called the recurrent or recursive case.

Recurrence Relation

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- To solve a recurrence relation $T(n)$ we need to derive a form of $T(n)$ that is not a recurrence relation.
 - ▣ Such a form is called a **closed form** of the recurrence relation.
- There are five methods to solve recurrence relations that represent the running time of recursive methods:
 - Iteration method (*unrolling and summing*)
 - Substitution method (*Guess the solution and verify by induction*)
 - Recursion tree method
 - Master theorem (Master method)
 - Using Generating functions or Characteristic equations

Recurrence Relation

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- To solve a recurrence relation $T(n)$ we need to derive a form of $T(n)$ that is not a recurrence relation.
 - ▣ Such a form is called a **closed form** of the recurrence relation.
- We discuss examples with
 - ▣ Decreasing function
 - ▣ Dividing function

MASTER THEOREM



MASTER THEOREM FOR DECREASING FUNCTION



Master Theorem

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Master Theorem for Decreasing Function

- $T(n) = T(n - 1) + c \quad \dots \quad O(n)$
- $T(n) = T(n - 1) + n \quad \dots \quad O(n^2)$
- $T(n) = T(n - 1) + n \log n \quad \dots \quad O(n^2 \log n)$

- $T(n) = 2T(n - 1) + c \quad \dots \quad O(2^n)$
- $T(n) = 3T(n - 1) + c \quad \dots \quad O(3^n)$
- $T(n) = 2T(n - 1) + n \quad \dots \quad O(n2^n)$

Master Theorem

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Master Theorem for Decreasing Function

$$T(n) = aT(n - b) + f(n)$$

where,

$$a > 0, b > 0, \text{ and } f(n) = O(n^k)$$

where,

$$k \geq 0,$$

Master Theorem

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Master Theorem for Decreasing Function (Case 1)

- $T(n) = T(n - 1) + c \quad \dots \quad O(n)$
- $T(n) = T(n - 1) + n \quad \dots \quad O(n^2)$
- $T(n) = T(n - 1) + n \log n \quad \dots \quad O(n^2 \log n)$

$$\text{If } a = 1, T(n) = O(n^{k+1})$$
$$T(n) = O(n \times f(n))$$

Master Theorem

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Master Theorem for Decreasing Function (Case 2)

$$\square T(n) = 2T(n-1) + c \quad \dots \quad O(2^n)$$

$$\square T(n) = 3T(n-1) + c \quad \dots \quad O(3^n)$$

$$\square T(n) = 2T(n-1) + n \quad \dots \quad O(n2^n)$$

$$\text{If } a > 1, T(n) = O(n^k a^{n/b})$$
$$T(n) = O(f(n) \times a^{n/b})$$

$$\square T(n) = 2T(n-2) + c \quad \dots \quad O(2^{n/2})$$

Master Theorem

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Master Theorem for Decreasing Function (Case 3)

$$\square T(n) = \frac{1}{2} T(n-1) + c \quad \dots \quad O(1)$$

$$\square T(n) = \frac{1}{3} T(n-1) + c \quad \dots \quad O(1)$$

$$\square T(n) = \frac{1}{5} T(n-1) + n \quad \dots \quad O(n)$$

If $a < 1$, $T(n) = O(n^k) = O(f(n))$

Master Theorem

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Master Theorem for Decreasing Function

□ If $a < 1$,

$$T(n) = O(n^k) = O(f(n))$$

□ If $a = 1$,

$$T(n) = O(n^{k+1})$$

$$T(n) = O(n \times f(n))$$

□ If $a > 1$,

$$T(n) = O(n^k a^{n/b})$$

MASTER THEOREM FOR DIVIDING FUNCTION



Master Theorem

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Master Theorem for Dividing Function

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

where,

$$a \geq 0, b > 1, \text{ and } f(n) = \theta(n^k \log^p n)$$

From this, we get two values,

- $\log_b a$
- k

Master Theorem

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Master Theorem for Dividing Function

Case 1: If $\log_b a > k$, then $\theta(n^{\log_b a})$

Case 2: If $\log_b a = k$

- $p > (-1)$, then $\theta(n^k \log^{p+1} n)$
- $p = (-1)$, then $\theta(n^k \log \log n)$
- $p < (-1)$, then $\theta(n^k)$

Case 3: If $\log_b a < k$

- $p \geq (0)$, then $\theta(n^k \log^p n)$
- $p < (0)$, then $O(n^k)$

Master Theorem

$$f(n) = \theta(n^k \log^p n)$$

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Master Theorem for Dividing Function ... Example 1

- $T(n) = 2T\left(\frac{n}{2}\right) + 1$
- $a = 2, b = 2$, and $f(n) = \theta(1) = \theta(n^0 \log^0 n)$
- $\log_2 2 = 1, k = 0, p = 0$
- $\log_b a > k, p = 0$

Case 1: If $\log_b a > k$, then $\theta(n^{\log_b a})$

- $\theta(n^{\log_2 2}) = \theta(n)$

Master Theorem

$$f(n) = \theta(n^k \log^p n)$$

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Master Theorem for Dividing Function ... Example 2

- $T(n) = 4T\left(\frac{n}{2}\right) + n$
- $a = 4, b = 2$, and $f(n) = \theta(n) = \theta(n^1 \log^0 n)$
- $\log_2 4 = 2, k = 1, p = 0$
- $\log_b a > k, p = 0$

Case 1: If $\log_b a > k$, then $\theta(n^{\log_b a})$

- $\theta(n^{\log_2 4}) = \theta(n^2)$

Master Theorem

$$f(n) = \theta(n^k \log^p n)$$

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Master Theorem for Dividing Function ... Example 3

- $T(n) = 9T\left(\frac{n}{3}\right) + n^2$
- $a = 9, b = 3$, and $f(n) = \theta(n^2) = \theta(n^2 \log^0 n)$
- $\log_3 9 = 2, k = 2, p = 0$
- $\log_b a = k, p = 0$

Case 2: If $\log_b a = k$, and

- $p > (-1)$, then $\theta(n^k \log^{p+1} n)$
- $\theta(n^2 \log n)$

Master Theorem

$$f(n) = \theta(n^k \log^p n)$$

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Master Theorem for Dividing Function ... Example 4

- $T(n) = 4T\left(\frac{n}{2}\right) + n^2 \log n$
- $a = 4, b = 2, \text{ and } f(n) = \theta(n^2 \log n)$
- $\log_2 4 = 2, k = 2, p = 1$
- $\log_b a = k, p = 1$

Case 2: If $\log_b a = k$, and

- $p > (-1)$, then $\theta(n^k \log^{p+1} n)$
- $\theta(n^2 \log^2 n)$

Master Theorem

$$f(n) = \theta(n^k \log^p n)$$

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Master Theorem for Dividing Function ... Example 5

- $T(n) = 4T\left(\frac{n}{2}\right) + \frac{n^2}{\log n}$
- $a = 4, b = 2, \text{ and } f(n) = \theta(n^2 \log^{-1} n)$
- $\log_2 4 = 2, k = 2, p = -1$
- $\log_b a = k, p = -1$

Case 2: If $\log_b a = k$, and

- $p = (-1)$, then $\theta(n^k \log \log n)$
- $\theta(n^2 \log \log n)$

Master Theorem

$$f(n) = \theta(n^k \log^p n)$$

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Master Theorem for Dividing Function ... Example 6

- $T(n) = T\left(\frac{n}{2}\right) + n^2$
- $a = 1, b = 2$, and $f(n) = \theta(n^2 \log^0 n)$
- $\log_2 1 = 0, k = 2, p = 0$
- $\log_b a < k, p = 0$

Case 3: If $\log_b a < k$, and

- $p \geq (0)$, then $\theta(n^k \log^p n)$
- $\theta(n^2)$

Master Theorem

$$f(n) = \theta(n^k \log^p n)$$

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Master Theorem for Dividing Function ... Example 7

- $T(n) = T\left(\frac{n}{2}\right) + \frac{n^2}{\log n}$
- $a = 1, b = 2, \text{ and } f(n) = \theta(n^2 \log^{-1} n)$
- $\log_2 1 = 0, k = 2, p = -1$
- $\log_b a < k, p = -1$

Case 3: If $\log_b a < k$, and

- $p < (0)$, then $O(n^k)$
- $O(n^2)$

SUBSTITUTION METHOD



Substitution Method

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- The most general method,
 1. **Guess** the form of the solution.
 2. **Verify** by induction.
 3. **Solve** for constants.
- **Example:** We can use the substitution method to establish either upper or lower bounds on a recurrence. As an example, let us determine an upper bound on the recurrence

$$T(n) = 2T(\lfloor n/2 \rfloor) + n ,$$

$$T(n) = 2T(\lfloor n/2 \rfloor) + n ,$$

Substitution Method

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We guess that the solution is

$$T(n) = O(n \lg n).$$

The substitution method requires us to prove that

$$T(n) \leq cn \lg n$$

for an appropriate choice of the constant $c > 0$.

We start by assuming that

$$T(\lfloor n/2 \rfloor) \leq c \lfloor n/2 \rfloor \lg(\lfloor n/2 \rfloor).$$

Substitution Method

$$\lg\left(\frac{n}{2}\right) = \lg(n) - \lg(2)$$

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$$T(n) = O(n \lg n).$$

$$T(n) \leq cn \lg n$$

$$T(\lfloor n/2 \rfloor) \leq c \lfloor n/2 \rfloor \lg(\lfloor n/2 \rfloor).$$

$$\begin{aligned} T(n) &= 2T(\lfloor n/2 \rfloor) + n, \\ T(n) &\leq 2(c \lfloor n/2 \rfloor \lg(\lfloor n/2 \rfloor)) + n \\ &\leq cn \lg(n/2) + n \\ &= cn \lg n - cn \lg 2 + n \\ &= cn \lg n - cn + n \\ &\leq cn \lg n, \end{aligned} \quad \boxed{c \geq 1.}$$

Acknowledgement

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Introduction to Algorithms

Thomas H. Cormen, Charles E. Leiserson,
Ronald L. Rivest, and Clifford Stein